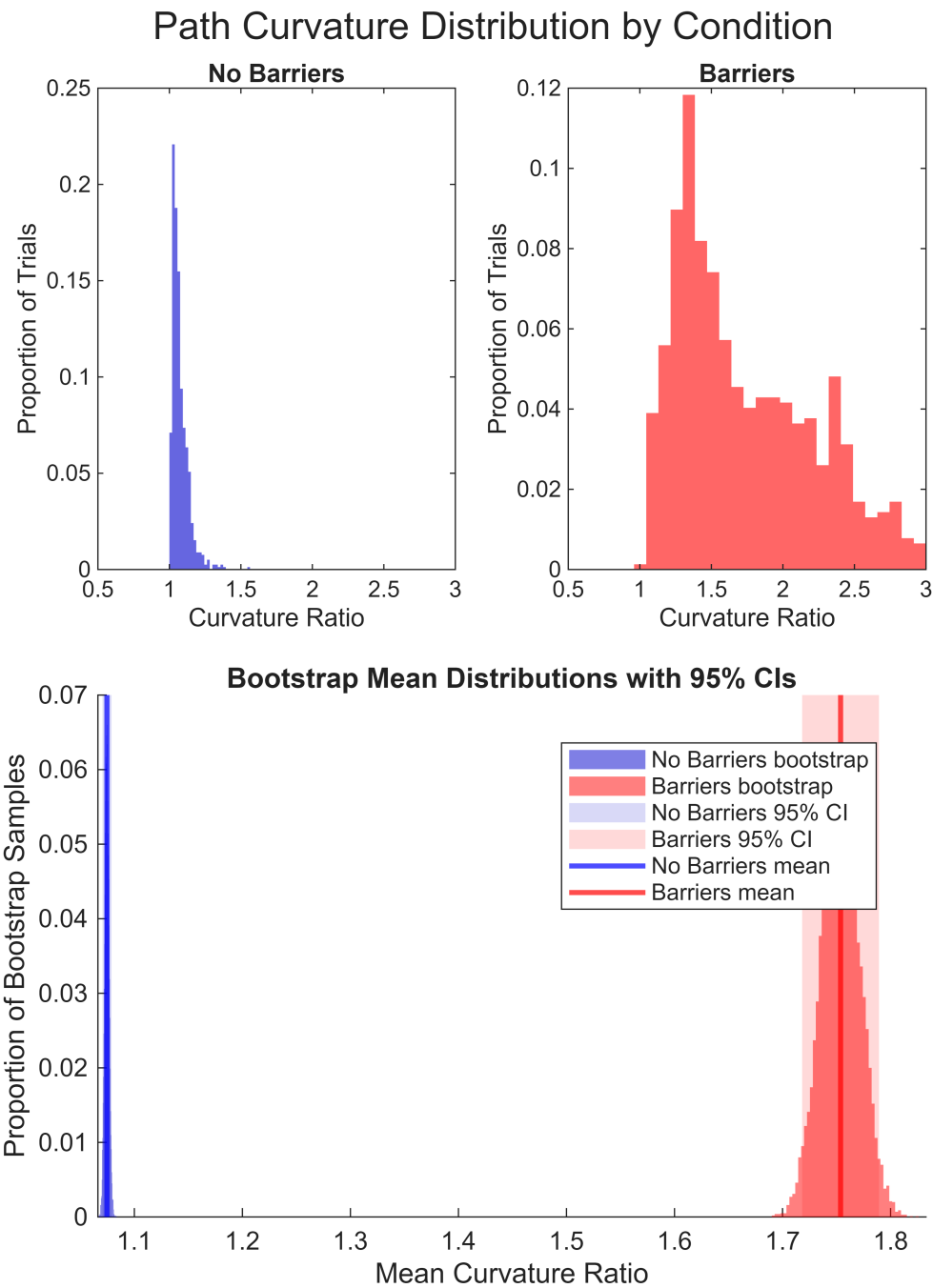


Final Project Writeup

Behavioral: Curvature Metric

Behavioral: Find a curvature metric (ratio of actual path length to straight line distance between start and end). Plot the distribution of curvature with a bootstrapped confidence interval for curved and straight path conditions. This would be a behavioral verification that the animal actually did the curved paths and not just the straight length ones.

```
curvatureMetric(R)
```



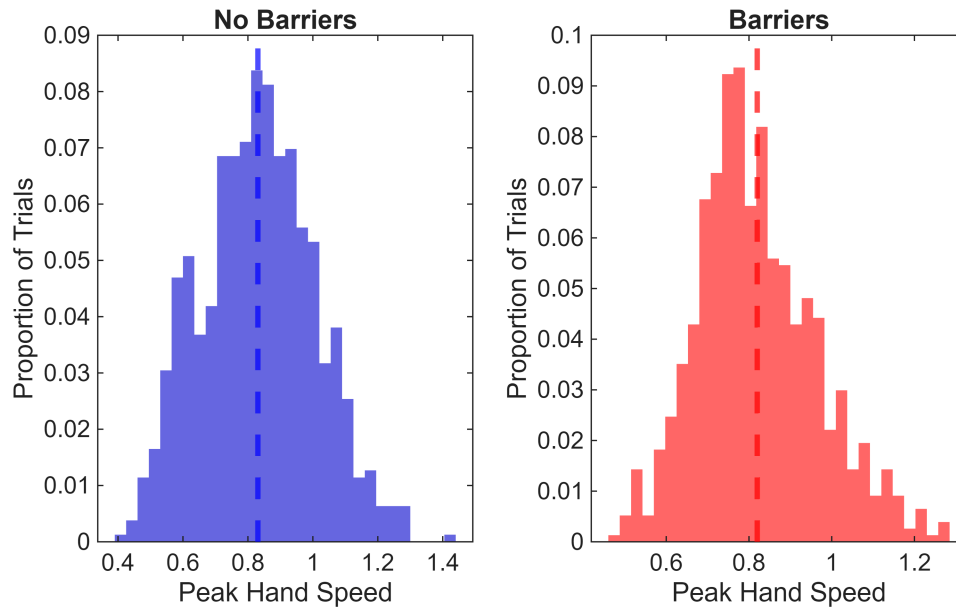
Behavioral: Peak Hand Velocity

Behavioral: Hand velocity over time split by the curved and straight reaches. Compute peak hand speed during trial, split trials into groups based on curved or straight reaches, compare the means.

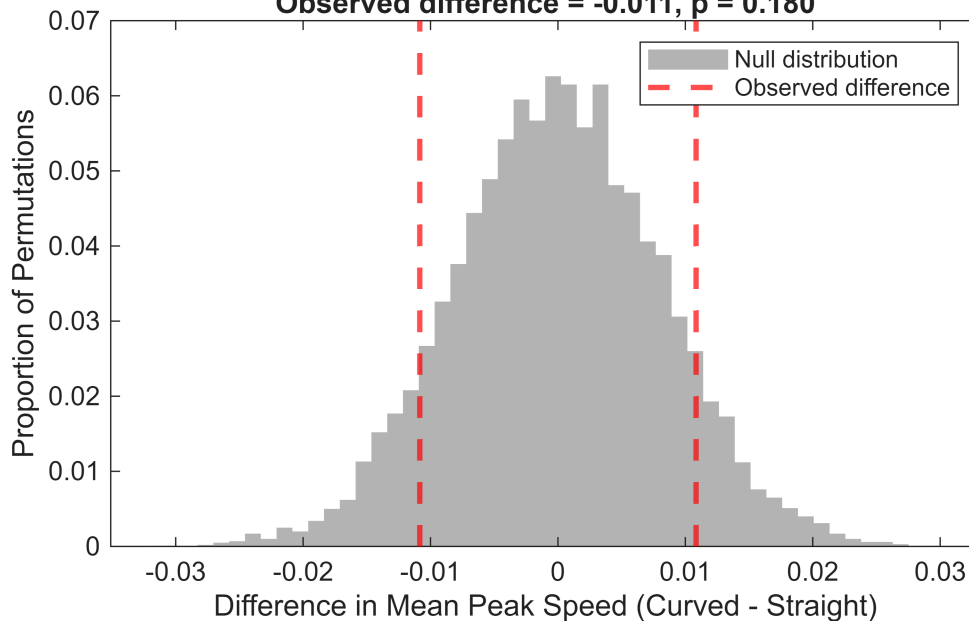
Monte Carlo: shuffle the curved/straight labels, recompute the difference in the means, do this 1,000 times to create a null distribution. P-value would show us if curved and straight reaches have significantly different peak hand speeds.

```
handVelocityMonteCarlo(R)
```

Peak Hand Speed Distribution by Condition



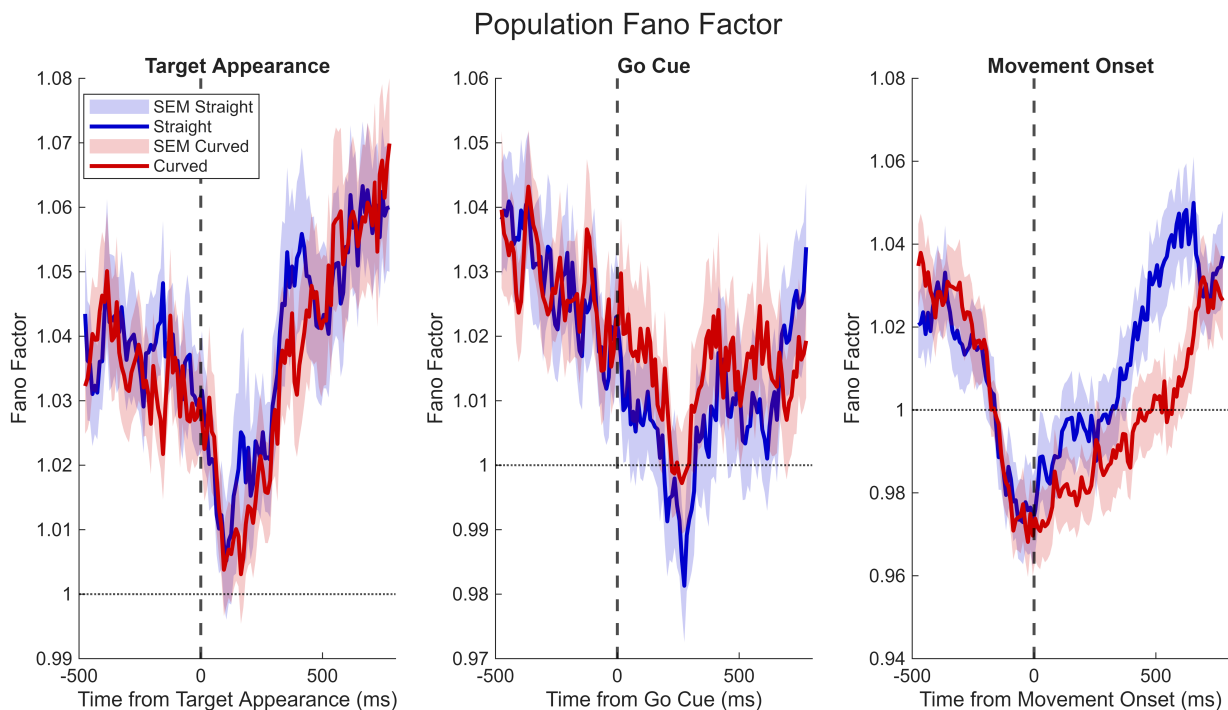
Null Distribution (Monte Carlo Permutation Test) Observed difference = -0.011, p = 0.180



Trial by Trial Variability

We will compute the Fano Factor as a function of time aligned to both the target appearance, movement onset, and go-cue separately for curved and straight reaches to determine how variability evolves across delay and movement and whether this differs for the reach types, considering curved reaches are arguably more complex.

```
trialToTrialVariability(R)
```



As we learned in class, the Fano Factor (FF) is used to quantify neural variability, with $FF=1$ indicating regular firing. The plot shown was averaged. At target appearance, the variability drops, and steadily rises. The same pattern is shown in the go cue alignment plot. When aligned to the movement onset, we see the straight reaches have a higher variability after movement onset compared to the curved reaches. This could mean that at movement onset, the curved trajectories execute movement in a stereotyped way. These results suggest that straight reaches are more locked to instruction, and curved reaches are more locked to execution. However, it is important to note that these differences in variability are minor. In addition, averaging over the population might minimize the differences in variability we see between the two conditions.

NOTE: Put in FF between arrays and single units

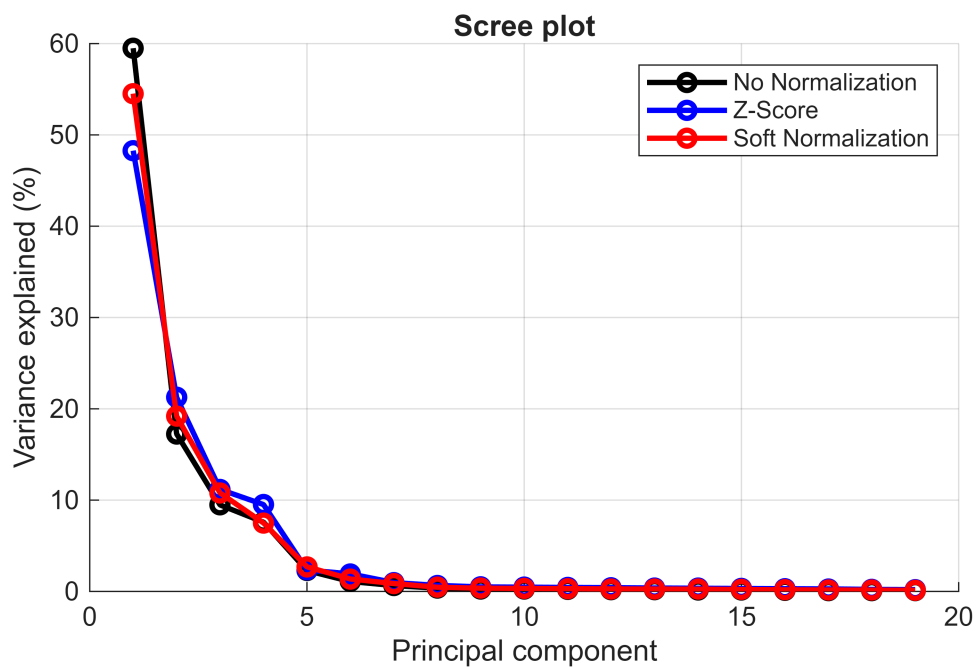
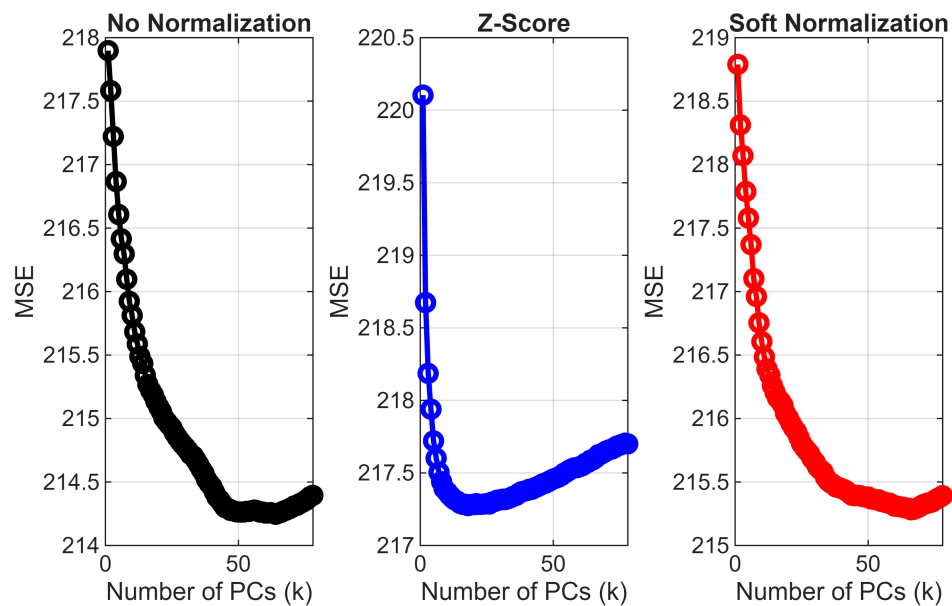
Linear Dimensionality Reduction

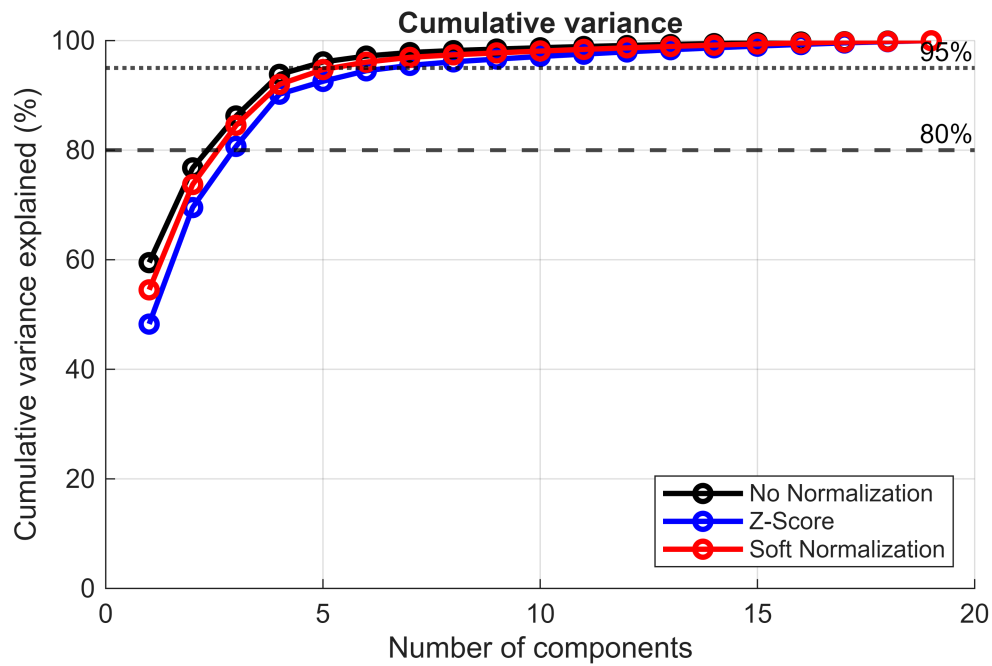
We will do cross-validated PCA on M1 activity to determine the true low dimensional structure of the neuron's responses, comparing dimensionality under no normalization, z-scoring, and soft normalization.

```
results = linearDimensionalityReduction(R);
```

Firing rates created.

Cross-validated PCA loss by dimensionality



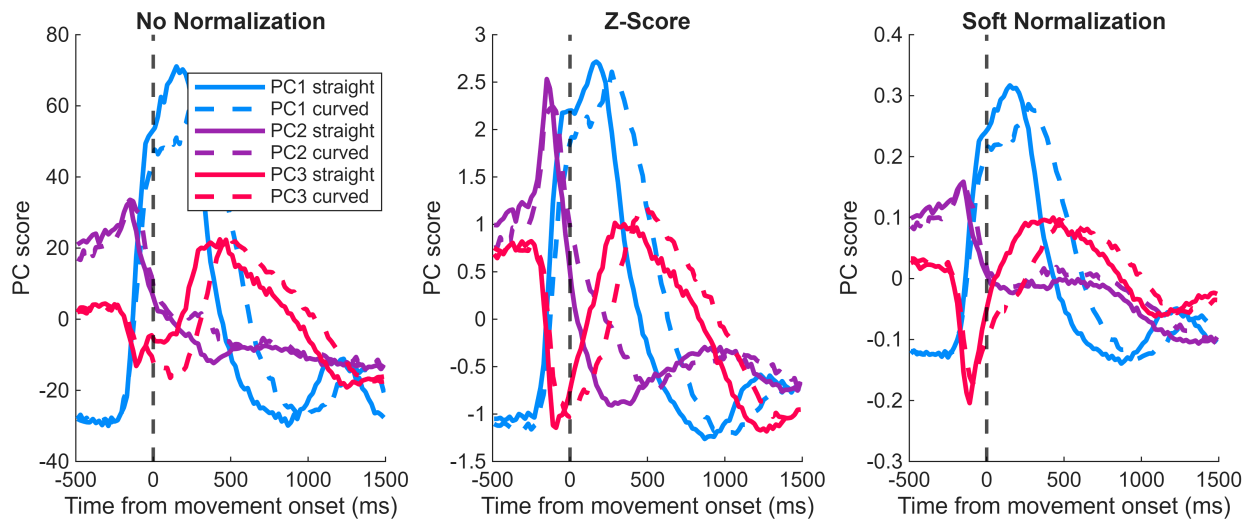


PCA Trajectory Plots

```
pcaTrajectories(R)
```

Firing rates created.

Top 3 PCs Over Time



In all of these PCA trajectories, we noticed that PC1 appears to peak at movement onset. Therefore, we believe that PC1 is representative of the actual movement execution that the animal does. This peak is preserved regardless of the normalization. Based on the feedback in class, we cannot say for sure what PC2 and PC3 encode for in movement phase. We also noticed that the curved trajectories show a slight lag when compared to the straight trajectories. This might be because the curved trajectories are more complex and require more online error correction during movement execution than the straight trajectories.

Classifier

We will classify curved vs. straight reaches based on neural data from the delay and movement period separately using SVM and logistic regression to assess when kinematic information is encoded in M1. We will assess the significance of our classifier using a Monte Carlo shuffle test, where we shuffle curved vs straight trial labels 1000 times to generate a null distribution and compute a p-value for each method/time-window combination.

```
classifier(R)
```

Firing rates computed.

Straight (No Barriers):

Value	Count	Percent
1	104	13.20%
2	133	16.88%
3	221	28.05%
4	152	19.29%
5	83	10.53%
6	95	12.06%

Curved (Barriers):

Value	Count	Percent
1	109	14.17%
2	119	15.47%
3	219	28.48%
4	146	18.99%
5	85	11.05%
6	91	11.83%

```
=====
SVM (RBF) | Delay Period | Straight (No Barriers)
=====
Observed accuracy = 0.280
p-value = 1.0000
```

```
=====
Logistic Regression | Delay Period | Straight (No Barriers)
=====
Observed accuracy = 0.942
p-value = 0.0010
```

```
=====
SVM (RBF) | Movement Period | Straight (No Barriers)
=====
Observed accuracy = 0.280
p-value = 1.0000
```

```
=====
Logistic Regression | Movement Period | Straight (No Barriers)
=====
Observed accuracy = 0.982
p-value = 0.0010
```

```
=====
SVM (RBF) | Delay Period | Curved (Barriers)
=====
Observed accuracy = 0.285
p-value = 1.0000
```

```
=====
Logistic Regression | Delay Period | Curved (Barriers)
=====
```

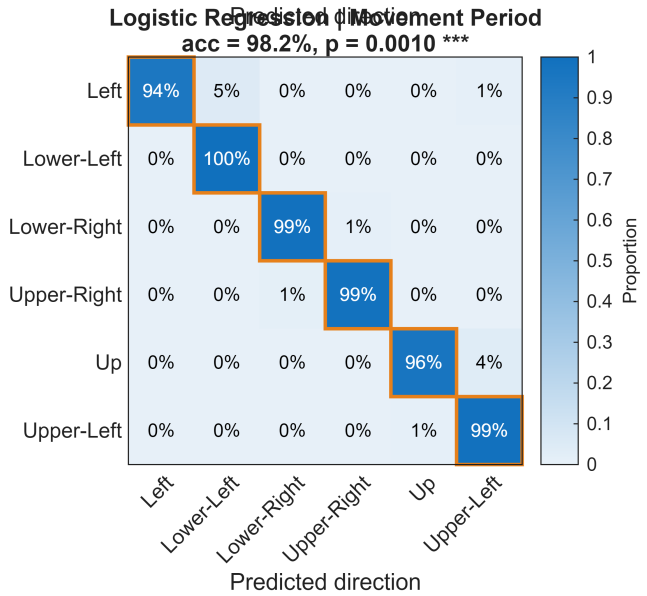
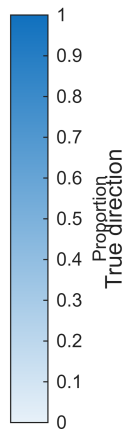
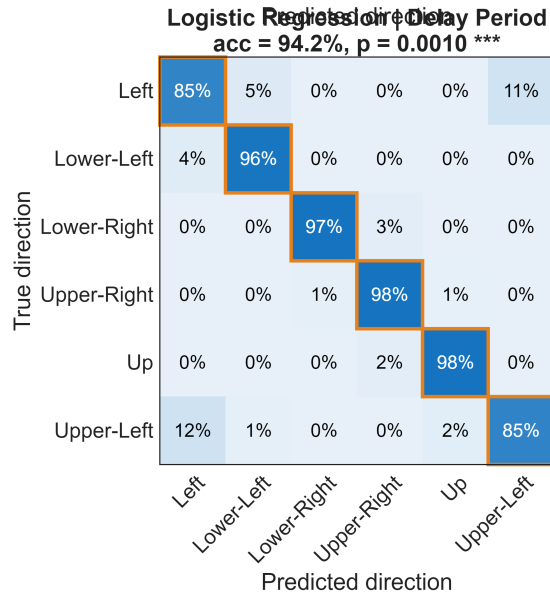
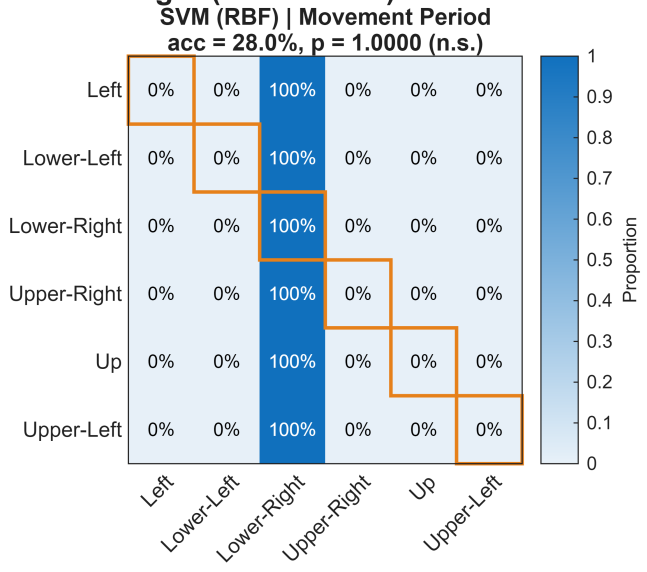
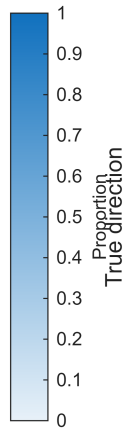
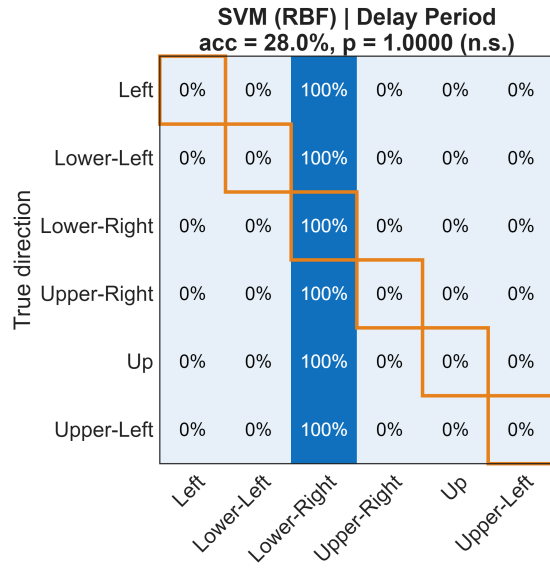
Observed accuracy = 0.828
p-value = 0.0010

=====
SVM (RBF) | Movement Period | Curved (Barriers)
=====
Observed accuracy = 0.285
p-value = 1.0000

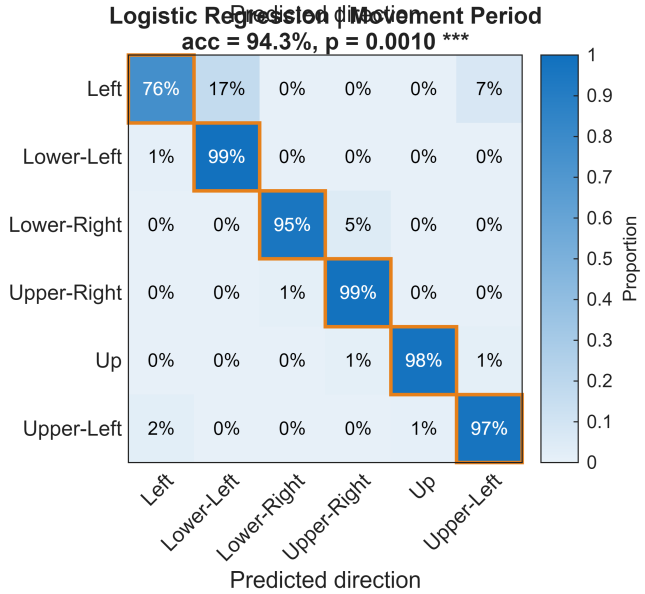
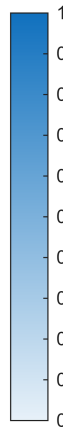
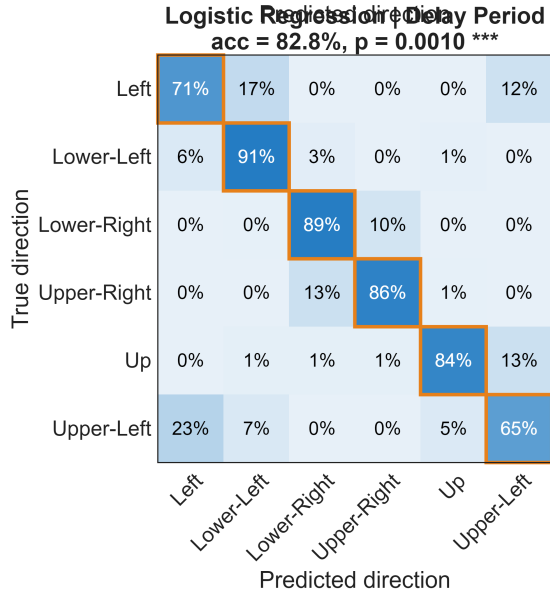
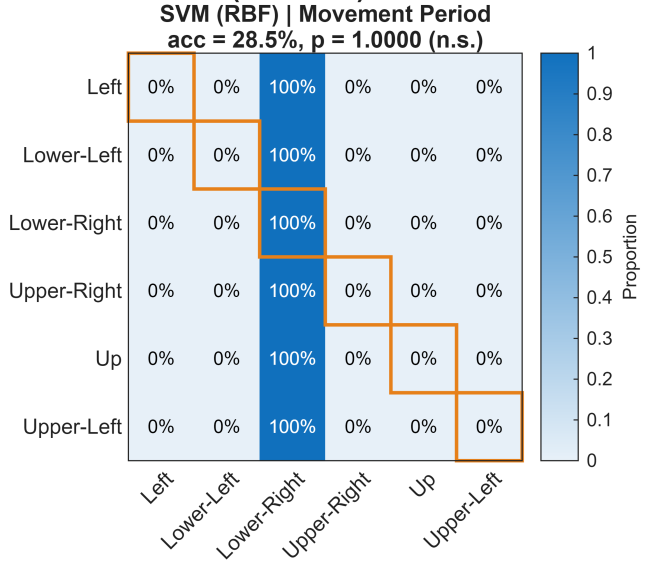
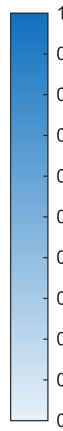
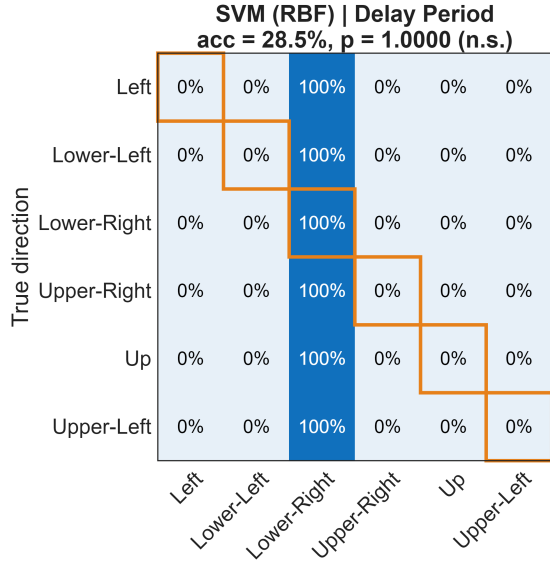
=====
Logistic Regression | Movement Period | Curved (Barriers)
=====
Observed accuracy = 0.943
p-value = 0.0010

Method	Window	Condition	Accuracy	p-value	
SVM (RBF)	Delay Period	Straight (No Barriers)	0.280	1.0000	(n.s.)
Logistic Regression	Delay Period	Straight (No Barriers)	0.942	0.0010	***
SVM (RBF)	Movement Period	Straight (No Barriers)	0.280	1.0000	(n.s.)
Logistic Regression	Movement Period	Straight (No Barriers)	0.982	0.0010	***
SVM (RBF)	Delay Period	Curved (Barriers)	0.285	1.0000	(n.s.)
Logistic Regression	Delay Period	Curved (Barriers)	0.828	0.0010	***
SVM (RBF)	Movement Period	Curved (Barriers)	0.285	1.0000	(n.s.)
Logistic Regression	Movement Period	Curved (Barriers)	0.943	0.0010	***

Direction Confusion Matrix — Straight (No Barriers)



Direction Confusion Matrix — Curved (Barriers)



```
ans = struct with fields:
  obs_acc: [2x2x2 double]
  perm_acc: [2x2x2x1000 double]
  p_values: [2x2x2 double]
  pred: {2x2x2 cell}
  true_lab: {2x2x2 cell}
  params: [1x1 struct]
```

In both the curved and straight trajectories, the classifiers performed better when trained on the movement period than on the delay period. This suggests that neural activity in the movement period more richly encodes directional information than in the delay period. In addition, both the SVM classifier with an RBF kernel and the logistic regression performed well. This might suggest that the directions are linearly separable in neural space, as both a linear and nonlinear classifier could successfully decode direction. Also, between the curved and straight trajectories, the classifier performed well. This suggests that even though the curved trajectory might have been a more complex behavior, direction is still separable.